

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **production ML pipelines** for enterprise data workflows using **Python, pandas**, and **scikit-learn**, automating data preparation, model training, and evaluation across **petabyte-scale** banking transaction datasets; reduced manual data wrangling time by **60%** and improved pipeline reliability to **99.8% SLA**.
- Built **LLM-based generative AI solutions** including **RAG systems** and **prompt engineering** implementations for enterprise use cases, integrating with **Apple's enterprise data platform** infrastructure via **REST APIs** and **SQL** data layer; delivered **3 production GenAI agents** serving **8 business units** with **92% user adoption**.
- Developed **automated ML pipelines** spanning data prep, training, evaluation, and deployment using **Jenkins CI/CD, Git, and Docker containerization**; integrated **model monitoring** dashboards in **Power BI** and **Dynatrace** to track real-world performance metrics, reducing model drift detection latency from **72 hours to 4 hours**.
- Applied **data science techniques** including **anomaly detection, forecasting**, and predictive analytics to solve real business problems in payments operations; leveraged **statistical methods, time-series analysis**, and **A/B testing** to drive **\$2.3M in operational cost savings** across enterprise finance workflows.
- Architected **distributed data systems** and **large-scale data processing** workflows using **AWS Lambda, S3**, and **PostgreSQL/Oracle** for processing **billions of rows daily**; optimized **ETL pipelines** and schema designs to reduce query latency by **45%** and support **petabyte-scale** analytics at enterprise scale.
- Collaborated with **cross-functional partners** and business stakeholders to scope, design, and ship **ML systems** solving real problems; partnered with enterprise BAs and operations teams on requirements gathering, UAT coordination, and **change-control management** for **7+ production deployments** with **zero post-go-live incidents**.

SKILLS

Machine Learning & Generative AI – LLMs, RAG systems, prompt engineering, agentic frameworks, generative AI solutions, model development, model evaluation, model monitoring.

Data Engineering & Pipelines – automated ML pipelines, data preparation, ETL, data wrangling, distributed data systems, large-scale data processing, schema design, data modeling.

Programming & Production Code – Python, Java Spring Boot, SQL, REST APIs, pandas, NumPy, scikit-learn, Flask, FastAPI, solid production code.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, CI/CD pipelines, Linux.

Business Intelligence & Analytics – Power BI, Tableau, DAX, Power Query, BI dashboards, data warehousing, anomaly detection, forecasting, statistical analysis, A/B testing, time-series analysis.

Databases & Data Platforms – PostgreSQL, MySQL, Oracle, enterprise data warehouse, petabyte-scale data systems, data modeling.

Enterprise & Domain Knowledge – enterprise data platform infrastructure, cross-functional collaboration, change-control management, UAT coordination, Agile/Scrum, Jira, Confluence.

EDUCATION AND TRAINING

PROJECTS

Enterprise Generative AI & Predictive Analytics Platform (HCL Technologies)

- Architected and deployed an **end-to-end ML pipeline** for enterprise-wide **generative AI and predictive analytics** on petabyte-scale datasets, integrating **LLM-based RAG agents**, **prompt engineering**, and **agentic frameworks** (LangChain patterns) to automate business intelligence workflows across **12 business units**.
- Built **automated data preparation and model training** infrastructure using **Python**, **pandas**, **scikit-learn**, and **AWS Lambda** to process **billions of rows daily**; orchestrated pipeline execution via **Jenkins CI/CD** with **Docker containerization**, achieving **99.8% uptime SLA** and **< 2-hour training cycles**.
- Developed **anomaly detection** and **forecasting models** to identify operational risks and predict enterprise demand; deployed models via **REST APIs** and integrated **model monitoring dashboards** in **Power BI** and **Dynatrace**, reducing incident response time by **65%** and enabling **\$2.3M in cost optimization**.
- Implemented **RAG-based knowledge retrieval systems** and **generative AI solutions** for enterprise use cases, integrating with **PostgreSQL** and **Oracle** data layers via **SQL** queries; validated model accuracy and real-world performance through rigorous **evaluation metrics** and **cross-functional UAT**.
- Collaborated with **enterprise stakeholders**, data scientists, and platform engineers to scope requirements, iterate on model behavior based on business feedback, and ship **4 production ML/GenAI systems** with **zero post-deployment incidents** and **88% stakeholder satisfaction**.

SWIFT Payments Reconciliation & Enterprise Data Integration Dashboard (HCL Technologies)

- Engineered **data wrangling and ETL pipelines** for **SWIFT MT/MX message flows** and **ISO 20022** transaction streams, processing **100M+ daily payment messages** across **SEPA** and cross-border payment corridors; automated data validation and reconciliation using **Python**, **SQL**, and **Oracle** to achieve **99.99% data accuracy**.
- Developed **BI dashboards** in **Power BI** with **DAX measures** and **Power Query ETL** logic to visualize real-time payments operations, transaction flows, and regulatory compliance metrics; published interactive reports for **50+ business stakeholders** reducing operational reporting overhead by **40%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)